



# Croatian Open Competition in Informatics

Round 2, November 22<sup>nd</sup> 2025

## Tasks

| Task               | Time limit | Memory limit | Score |
|--------------------|------------|--------------|-------|
| <b>Minesweeper</b> | 5 seconds  | 512 MiB      | 50    |
| <b>Dodatna</b>     | 1 second   | 512 MiB      | 70    |
| <b>Natjecanje</b>  | 1 second   | 512 MiB      | 110   |
| <b>Rastući</b>     | 1 second   | 512 MiB      | 110   |
| <b>Tornjevi</b>    | 1 second   | 512 MiB      | 110   |
| <b>Total</b>       |            |              | 450   |

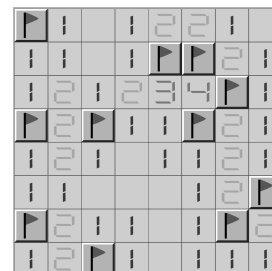


## Task Minesweeper

Marko is a regular university student who still shows up to every lecture, even though his stomach and head often ache from studying late into the night. Because he's also a gifted student, he often finds the lectures boring, so instead of listening he plays his favorite game, *Minesweeper*. Eventually, even that got boring, so he decided to create his own version of the game.

In Marko's version of *Minesweeper*, he starts the game already knowing the positions of all  $k$  bombs on a board with  $n$  rows and  $m$  columns. Each cell can contain at most one bomb.

His task is to fill every non-bomb cell with the number of bombs in the eight surrounding cells. Cells that contain bombs are marked with "B". Your task is to output the matrix generated according to Marko's rules.



### Input

In the first line are the natural numbers  $n$ ,  $m$  and  $k$  ( $1 \leq n, m \leq 500, 1 \leq k \leq n \cdot m$ ), the numbers from the problem statement.

In each of the following  $k$  lines are the numbers  $r_i$  and  $s_i$  ( $1 \leq r_i \leq n, 1 \leq s_i \leq m$ ), the row and column in which one of the bombs is located.

### Izlazni podaci

In  $n$  rows, output  $m$  characters separated by a space, the matrix produced according to the rules of Marko's game.

### Scoring

| Subtask | Points | Constraints                |
|---------|--------|----------------------------|
| 1       | 15     | $n = 1$                    |
| 2       | 18     | $k = 1$                    |
| 3       | 17     | No additional constraints. |

### Examples

#### input

1 6 1

1 3

#### output

0 1 B 1 0 0

#### input

3 3 3

1 1

2 3

1 3

#### output

B 3 B

1 3 B

0 1 1

**Clarification of the first example:** In the cell in the 3rd column there is a bomb (the letter "B"). The cells of the matrix in the 2nd and 4th columns (marked with the number 1) are adjacent to it.



## Task Dodatna

The famous mentor Jakov loves teaching and helping diligent students. This school year he decided to help some truly unusual students and teach additional computer science classes. Namely, each of the  $n$  students has very strange requirements that must be satisfied for them to even show up to the extra class, but more on that another time.

Jakov's conditions for holding the extra class are much simpler and are as follows:

- All students who will attend the extra class must be **in the school from the beginning to the end** of the extra class.
- There must be at least  $k$  students in the extra class, otherwise it will not be held at all.



With the help of the principal, Jakov discovered the schedules of all  $n$  students, and for each student  $i$  ( $1 \leq i \leq n$ ) he knows that they are in the school between the  $l_i$ -th and  $r_i$ -th millisecond of the day (exclusively\*).

Help Jakov by answering the question: what is the maximum possible duration of the extra class he can hold? If Jakov cannot hold an extra class at all, output the number 0.

\*The first millisecond during which student  $i$  is in the school is  $l_i$ , and the last is  $r_i - 1$ .

### Input

The first line contains the natural numbers  $n$  and  $k$  ( $1 \leq n, k \leq 3 \cdot 10^5$ ) from the problem statement.

Each of the following  $n$  lines contains 2 numbers  $l_i$  and  $r_i$  ( $1 \leq l_i < r_i \leq 86\,400\,000$ ), from the problem statement.

### Output

In the first and only line, output a single number — the maximum duration of the extra class that Jakov can organize, or 0 if it is impossible.

### Scoring

| Subtask | Points | Constraints                 |
|---------|--------|-----------------------------|
| 1       | 13     | $K = 1$                     |
| 2       | 27     | $1 \leq N \leq 1000, K = 2$ |
| 3       | 11     | $r_i \leq 100$              |
| 4       | 19     | No additional constraints.  |



## Examples

**input**

5 1  
1 3  
1 4  
1 5  
1 6  
1 7

**output**

6

**input**

5 2  
6 10  
8 14  
5 9  
5 6  
4 6

**output**

3

**Clarification of the second example:** The longest duration of the extra class that Jakov can organize is 3 seconds because the 1st and 3rd student will attend it in the 6th, 7th, and 8th second.



## Task Natjecanje

Dino has been practicing for a package delivery competition for weeks. In the competition, the winner is the contestant who delivers all the packages from the warehouse to the designated destinations the fastest. Unfortunately, Dino has only two hands, so he can carry at most two packages at a time.

He can drop and pick up packages however he wants, as long as he carries at most two at a time. He can also carry and deliver them in any order, and each package can be delivered to any destination. After completing his delivery route, there must be a package at each of the  $k$  destinations.

During the competition, Dino moves on a grid of  $n$  rows and  $m$  columns, where free cells are marked with ".", obstacles with "#", Dino's starting cell and all packages with "S", and package destinations with "X". In one second, he can move to any adjacent cell in the four directions (up, down, left, or right) that is not an obstacle. Dropping and picking up packages takes negligible time.

Help Dino determine the minimum number of seconds he needs to deliver all packages to their destinations and return to the starting cell, or print -1 if it is impossible.



### Input

In the first line, there are three natural numbers  $n$ ,  $m$ , and  $k$  ( $1 \leq n, m \leq 500, 1 \leq k \leq 67$ ), the number of rows and columns of the matrix and the number of package destinations.

In each of the next  $n$  lines, there are  $m$  characters  $c_{ij}$  describing the board. Each character of the matrix will be one of ".", "#", "S", or "X" (without quotes).

The letter "X" will appear exactly  $k$  times in the matrix.

### Output

In the first line, print a single number - the minimum number of seconds required for Dino to deliver all the packages to their destinations and return to the starting cell, or -1 if this is impossible.

### Scoring

| Subtask | Points | Constraints                |
|---------|--------|----------------------------|
| 1       | 17     | $k = 2$                    |
| 2       | 26     | $k \leq 16$                |
| 3       | 29     | $k \leq 22$                |
| 4       | 38     | No additional constraints. |



## Examples

**input**

```
5 5 3
X...X
.....
.....
.....
S...X
```

**output**

24

**input**

```
5 5 4
..X..
#X#..
#...X
.SX#.
.....
```

**output**

16

**Clarification of the first example:** Dino will go with 1 package to the bottom-right destination, leave the package, and return to the cell with the letter S. Then he will take 2 packages and visit the top-right destination and the top-left destination. After that, he will return to the starting cell and finish his route in 24 seconds.



## Task Rastući

The European Union recently passed a new law stating that all sequences must be increasing, and starting from January 1, 2026, anyone possessing a sequence that is not increasing will be strictly penalized. More precisely, a sequence  $a_1, a_2, \dots, a_n$  is **illegal** if there exists  $i$  ( $1 \leq i < n$ ) such that  $a_i > a_{i+1}$ , and otherwise the sequence is **legal**.

Ivica recently received a sequence as a gift and is worried that he might get into trouble because of the new law. Fortunately, Ivica realized that he can take any two adjacent elements of his sequence and replace them with their sum. More precisely, if Ivica's sequence is currently  $a_1, a_2, \dots, a_m$ , he can choose some  $k$  such that  $1 \leq k < m$  and replace his sequence with  $a_1, a_2, \dots, a_{k-1}, (a_k + a_{k+1}), a_{k+2}, \dots, a_m$ . Ivica can perform this operation as many times as he wants, and his goal is to obtain a legal sequence using such operations.



Of course, Ivica does not want to completely destroy his sequence, so he would like to obtain the longest possible legal sequence from his initial sequence. Help Ivica determine the length of the longest legal sequence he can get from the initial sequence and output any such legal sequence of maximum length.

### Input

In the first line, there is a natural number  $n$  ( $1 \leq n \leq 5000$ ), the length of Ivica's sequence.

In the second line, there are  $n$  natural numbers  $a_i$  ( $1 \leq a_i \leq 10^9$ ), the sequence that Ivica received as a gift.

### Output

In the first line, print a single number  $m$  - the length of the longest valid sequence that Ivica can obtain from the initial sequence.

In the second line, print  $m$  numbers, the elements of one of the valid sequences of length  $m$  that Ivica can obtain from the initial sequence. If there are multiple such sequences, print any one of them.

### Scoring

| Subtask | Points | Constraints                |
|---------|--------|----------------------------|
| 1       | 10     | $n \leq 20$                |
| 2       | 15     | $n \leq 100, a_i \leq 100$ |
| 3       | 20     | $n \leq 500$               |
| 4       | 25     | $n \leq 1000$              |
| 5       | 40     | No additional constraints. |

60% of the points for a test case are earned by having a correct output of the 1st line.

The remaining 40% of the points for the test case are earned by having a correct output of the 2nd line.



## Examples

**input**

6  
3 2 6 3 3 8

**output**

4  
5 6 6 8

**input**

7  
3 6 4 2 6 2 5

**output**

5  
3 6 6 6 7

**Clarification of the first example:** In the first operation, Ivica will replace the first 2 elements of the array with their sum and obtain the illegal array [5, 6, 3, 3, 8]. Then, he will replace the third and fourth elements of the array with their sum and obtain the legal array [5, 6, 6, 8]. It can be shown that this is a legal array of maximal length that can be obtained from the initial array.



## Task Tornjevi

Dominik is truly a special student and he constantly enjoys playing with blue and red cubes of different sizes.

He decided to build towers while following two peculiar rules:

- A cube directly above another cube can never be the same color as that cube.
- All cubes in the tower above some cube must be smaller in size than that cube.



Dominik opened his toy box which contains exactly  $n$  cubes. For each  $i$  ( $1 \leq i \leq n$ ), Dominik has exactly one cube of size  $i$ , and it is either red or blue.

Dominik got bored of stacking towers, so he came up with  $q$  difficult questions. In each question, Dominik thinks of two numbers  $l$  and  $r$ . He wants to know the **minimum** number of towers he must build so that they contain only cubes whose sizes are between  $l$  and  $r$  (inclusive), and such that every tower follows his peculiar rules. Every cube of a size between  $l$  and  $r$  must be used, i.e., placed in some tower.

Dominik has to study a lot because it's time for midterms at the university, so he asks you for help in answering his  $q$  questions.

### Input

In the first line, there are two natural numbers  $n$  and  $q$  ( $1 \leq n, q \leq 1 \cdot 10^5$ ), the number of cubes and the number of questions.

In the second line, there is a string  $s$ , where the character  $s_i$  represents the color of the cube of size  $i$ . Each character of the string will be either "P" or "C".

In each of the following  $q$  lines, there are two numbers  $l_i$  and  $r_i$  ( $1 \leq l_i \leq r_i \leq n$ ), the left and right boundaries of Dominik's query.

### Output

In  $q$  lines, print the answers to Dominik's questions from the problem statement.

### Scoring

| Subtask | Points | Constraints                          |
|---------|--------|--------------------------------------|
| 1       | 23     | $1 \leq n, q \leq 10$                |
| 2       | 38     | $1 \leq n, q \leq 1000$              |
| 3       | 25     | There will be at most 20 blue cubes. |
| 4       | 24     | No additional constraints.           |



## Examples

**input**

7 4  
PPCPPCC  
1 7  
1 5  
3 7  
4 5

**output**

3  
3  
2  
2

**input**

6 2  
CCCCC  
1 6  
2 5

**output**

6  
4

**input**

16 1  
PPPCPCCCCCPPPPP  
1 16

**output**

6

**Clarification of the second example:** All the cubes are red, so any tower that contains at least 2 cubes does not follow Dominik's rules. It follows that each tower contains exactly 1 cube, and the answer to each question is the number of cubes between  $l$  and  $r$ , i.e.,  $r - l + 1$ .